Cereal 6 — Results and Findings

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The monolithic model with per-label thresholds is the maize result to quote: macro-F1 0.468 over 5 folds. Splitting the problem into five specialists made it worse (0.283), and per-label thresholds were the single most useful change.

Headline numbers (5-fold OOF)

Pipeline	Thresholds	Macro-F1	Micro-F1	Hamming loss
Monolithic	Tuned per label	0.468	0.543	0.204
Monolithic	0.5	0.432	0.492	0.264
Specialists (fold 1 only)	—	0.334	0.356	—
Specialists (all folds)	Tuned	0.283	—	—
Specialists (all folds)	0.5	0.260	—	—

Dashes are values not recorded. Tuned scores were chosen and scored on the same OOF predictions, so they are slightly optimistic.

Findings

- **Threshold tuning matters most.** Moving from 0.5 to per-label thresholds added 3.6 points of macro-F1 and cut Hamming loss by 6 points. Weighted BCE pushes rare labels toward high recall and low precision; the thresholds pull them back.
- **Gray leaf spot is easy, southern rust is not.** Per-label F1 was only recorded approximately: about 0.83 for GLS (1,084 positives) and about 0.14 for SR (39 positives). SR shows some signal but too few examples to learn from. Re-run the evaluation cell for exact per-label values.
- **Specialists lost.** Fold 1 was already 13 points of macro-F1 below the monolithic model, and the full run ended 18.5 points below. Likely causes are listed in the Experiments doc; neither has been tested.
- **Fine-tuning hurt S5.** Its frozen backbone gave the best validation AUC (0.8026); both unfreezing phases were lower.
- **This is a first benchmark.** No earlier deep-learning result exists for this 8-label field dataset, so these numbers set the baseline rather than compete with one.